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1. Administrative & Study Identification

Full Study Title: A Phase I study of IDH305 in patients with advanced malignancies that harbor IDH1R132 mutations **Short Title/ Acronym:** IDH305 Phase I Trial **Protocol Number:** CIDH305X2101 **NCT Number:** NCT02381886 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** IDH305 **Phase of Development:** Phase I **Medical Monitor:** Novartis Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To identify the maximum tolerated dose (MTD) and/or recommended Phase 2 dose (RP2D) of IDH305 in patients with advanced malignancies that harbor IDH1R132 mutations. **Secondary Objectives:** To evaluate the safety, tolerability, pharmacokinetics (PK), and preliminary anti-tumor activity (Overall Response Rate, Complete Remission) of IDH305, as well as to assess pharmacodynamic target engagement via reduction in 2-hydroxyglutarate (2-HG) levels.

2.2 Background & Rationale Isocitrate dehydrogenase enzyme 1 (IDH1) mutations at the 132nd amino acid residue (R132*) result in the cellular accumulation of the oncometabolite, 2-hydroxyglutarate (2-HG). IDH305 is an orally bioavailable, brain-penetrant, mutant-selective allosteric IDH1 inhibitor demonstrating target engagement in preclinical models. This first-in-human study was designed to identify the recommended dose for expansion of IDH305 in patients with advanced IDH1R132-mutant malignancies, including acute myeloid leukemia (AML), myelodysplastic syndrome (MDS), and advanced solid tumors (like gliomas).

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation and expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: 166 (Overall study estimated enrollment) **Number of Sites:** Multicenter (Global, including sites in the US, Australia, Canada, and Singapore) **Estimated Study Start:** March 2015 **Estimated Primary Completion:** November 2017

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Documented IDH1R132-mutant tumors (advanced malignancies including AML, MDS, glioma). 2. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2. 3. Adequate hematologic, renal, and hepatic organ function.

4.2 Exclusion Criteria 1. Patients who have received prior treatment with a mutant-specific IDH1 inhibitor (with the exception of glioma patients). 2. Medical conditions that would prevent the patient's participation due to safety concerns or compliance, such as clinically significant cardiac, respiratory, gastrointestinal, renal, hepatic, or neurological disease. 3. Acute Promyelocytic Leukemia. 4. Women who are pregnant or lactating.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: IDH305 administered ranging from 75 mg to 750 mg twice daily. **Route of Administration:** Oral **Duration of Treatment:** Continuous treatment until disease progression, unacceptable toxicity, or premature study halt.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. **Prohibited:** Prior treatment with mutant-specific IDH1 inhibitors (except in glioma), as well as concurrent antineoplastic therapy.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Mutational Testing (IDH1R132), Medical History | | **Baseline** | Day 0 | First Dose, PK/PD Sampling, Vital Signs | | **Interim** | Weekly / Cycle specific | Safety Labs (Hepatic monitoring), PK profiling, Bone Marrow/Imaging Efficacy Assessment | | **End of Study** | Progression / Premature Halt | Final Vital Signs, Exit Interview, IP Accountability |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D) determined by the incidence of Dose-Limiting Toxicities (DLTs). **Measurement Method:** Clinical observation and laboratory assessments (e.g., hyperbilirubinemia, ALT/AST levels) during the first treatment cycle.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetic parameters (Cmax, AUC, T1/2) and target engagement (reduction in

2-HG concentration). * Clinical Efficacy: Complete Remission (CR), CR with incomplete count recovery, and Overall Response Rate (ORR).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical and laboratory monitoring, with special attention to hepatotoxicity (increased blood bilirubin, ALT/AST elevations), differentiation syndrome, and tumor lysis syndrome. **Serious Adverse Events (SAEs):** Evaluated per standard ICH-GCP 24-hour reporting guidelines. Grade 3/4 AEs closely tracked. **Data Monitoring Committee (DMC):** Utilized a Bayesian hierarchical model with overdose control principles to evaluate DLTs and guide dose-escalation decisions.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who received at least one dose of IDH305). **Sample Size Justification:** Dose escalation modeled using a Bayesian hierarchical model with an overdose control principle to securely determine the MTD. **Handling of Missing Data:** Handled via standard descriptive statistics without formal imputation in dose-escalation phase analyses.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring during the dose-escalation Phase I protocol to monitor real-time dose-limiting toxicities. **Audit Trail:** eCRF locking and rigorous documentation of dose modifications (especially regarding hepatotoxicity management).

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11. Study Identifiers & Governance

Primary ID: CIDH305X2101 **Registry Number:** NCT02381886 **Sponsor/ Lead Organization:** Novartis Pharmaceuticals **Study Title:** A Phase I study of IDH305 in patients with advanced malignancies that harbor IDH1R132 mutations. **Official Regulatory Title:** A Phase I study of IDH305 in patients with advanced malignancies that harbor IDH1R132 mutations.

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12. Clinical Framework (Oncology / IDH1 Specific)

Primary Condition: Advanced Malignancies (Acute Myeloid Leukemia, Myelodysplastic Syndrome, Glioma) **Diagnosis Threshold:** Confirmed IDH1R132 mutation (e.g., IDH1R132C, IDH1R132H) **Medical Methodology:** Targeted Mutant-Selective Allosteric Inhibitor **Optimization Target:** Target engagement (2-HG concentration reduction) and defining the therapeutic window

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13. Study Procedures & Methodology

Management Pathway: Dose escalation and safety monitoring for Phase I oncology patients **Education Protocol (HCP):** AE/DLT reporting training (specifically regarding hepatotoxicity and differentiation syndrome) **Education Protocol (Patient):** Informed consent, daily oral dosing diaries, AE self-reporting **Quality Audit Mechanism:** Safety Review Committee dose escalation meetings and continuous PK/PD parameter reviews

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14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression or premature study termination (due to narrow therapeutic window) **Onsite Visit Cadence:** Frequent cycle 1 visits (Day 1, Day 8, Day 15, etc.) for close PK/PD and DLT monitoring **Remote Monitoring Frequency:** Standard safety check-ins **Checklist Review Interval:** End of each treatment cycle for dose-escalation/safety clearance

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15. Sign-off & Approval

| Role | Name | Signature | Date | | :--- | :--- | :--- | :--- | | **Principal Investigator** | [Name] | ____ **| [DD-MMM-YYYY] | | Clinical Operations Lead | [Name] | ___ |** ***[DD-MMM-YYYY]*** **|** **|** ***Lead Statistician*** **|** ***[Name]*** **|** _____ | [DD-MMM-YYYY] |